

Clone the repository (git clone YOUR_REPO_URL) before proceeding.

Zen Sweep — Software Instructions

For a judge who has never seen this project. **Zen Sweep is an Android digital-wellbeing app.** Blocking and mood overlays require a **custom Expo Dev Client** (not Expo Go). There is **no backend, no login, and no .env configuration.**

The runnable app lives in `frontend/`.

1. Install prerequisites

Install these **before** `npm install`.

1.1 Required (Android demo)

Tool	Version / notes
Git	Any recent version
Node.js	20.x LTS or 22.x LTS (needed for Expo SDK 54 / React Native 0.81)
npm	Comes with Node (lockfile is <code>frontend/package-lock.json</code>)
Android Studio	Latest stable. Use its bundled JDK (JBR / Java 17)
Android SDK	From Android Studio SDK Manager: Platform 36, Build-Tools 36.0.0, Platform-Tools (adb), NDK 27.1.12297006 (Expo will request this on first native build)
Physical Android phone	Strongly recommended. API 24+ (Android 7). Package name: <code>com.zensweep.app</code> . Overlay + Usage Access are unreliable or incomplete on many emulators.

Confirm:

```
node -v
npm -v
adb version
```

Set `JAVA_HOME` to Android Studio's JBR, for example:

- Windows: `C:\Program Files\Android\Android Studio\jbr`
- macOS: `/Applications/Android Studio.app/Contents/jbr/Contents/Home`

1.2 Optional

Tool	Why
Xcode (macOS only)	<code>npm run ios</code> exists, but the native module <code>zen-app-usage</code> is Android-only . iOS will not block WhatsApp or show mood overlays.
Expo Go	Do not use. Custom native code will not load.

1.3 Phone setup

1. Enable **Developer options** → **USB debugging**.
2. Connect USB. Accept the RSA prompt.
3. Confirm:

```
adb devices
```

You should see `device` , not `unauthorized` or `offline` .

2. Install dependencies

From the **repository root**:

```
cd frontend
npm install
```

This installs Expo SDK 54, React Native 0.81.5, NativeWind, React Navigation, AsyncStorage, and the local module `zen-app-usage` (`file:./modules/zen-app-usage`).

Do **not** skip `npm install` . The native module is not published to npm.

3. Environment variables / config files

None required.

- There is **no** `.env.example` .
- `frontend/.env` is empty and unused.
- `frontend/src/config/env.ts` is a placeholder (`export {}`).
- `Firebase / auth` folders under `src/services/firebase/` and `src/store/` are stubs, not wired up.

You can skip any env-file step.

4. Build and run the app

4.1 Physical Android phone (the path that actually demos blocking)

Keep the USB cable connected.

```
cd frontend
npx expo run:android
```

Equivalent script: `npm run android` .

What this does:

1. Compiles the Kotlin module and a **debug Dev Client APK**.
2. Installs `com.zensweep.app` on the device.
3. Starts Metro (often on **8081**; this project has also been run on **8083**).

If the phone cannot reach Metro over Wi-Fi (common on restricted networks), reverse the port (use the port Metro printed):

```
adb reverse tcp:8081 tcp:8081
adb reverse tcp:8083 tcp:8083
```

Then open the installed **Zen Sweep** app. If you see the Expo Dev Client URL screen, connect to:

```
http://127.0.0.1:8081
```

(or `8083` if that is what Metro printed)

Reload JS later with:

```
cd frontend
npx expo start --port 8081
```

Native overlay / service changes require `npx expo run:android` **again**. JS-only UI changes hot-reload.

4.2 Android emulator

You can install the Dev Client the same way (`npx expo run:android` with an AVD running). **Usage Access and “Display over other apps” often do not work well in the emulator**, so timer/mood overlays may never appear. Use a real phone for judging the core feature.

4.3 iOS

```
cd frontend
npm run ios
```

This only makes sense on macOS with Xcode. **App blocking and mood overlays will not work** (`modules/zen-app-usage/expo-module.config.json` `platforms: android only`).

4.4 Web (not a valid demo)

```
cd frontend
npm run web
```

Web cannot access UsageStats or system overlays. Do not judge blocking from the browser.

4.5 Grant permissions on the phone (required)

After first launch, complete onboarding, then:

1. Usage access

Settings → Apps → Special access → **Usage data access** → enable **Zen Sweep**.

The app also prompts this when you tap **Start Block**.

2. Display over other apps

Settings → Apps → Special access → **Appear on top** (or Display over other apps) → enable **Zen Sweep**.

3. **Notifications** (Android 13+)

Allow so the “Zen Sweep watching / block active” foreground-service notification can show.

4. Samsung / OEM: if the OS suggests **Deep sleep** or battery restriction after crashes, set Zen Sweep to **Unrestricted** / never sleep, or overlays will stop when you leave the app.

5. Login, credentials, and how to use the features

No accounts. No test users. No seed database.

All data is local **AsyncStorage** on the device.

5.1 First launch (onboarding)

1. Open **Zen Sweep**.
2. Welcome → **Start growing — it's free** (the **Sign in** button is not a real login; it continues the same setup).
3. Enter a **name** (required). Age optional.
4. Enter a **dream / goals**.
5. Pick **interests**.
6. Pick **trap apps** (WhatsApp, Instagram, YouTube, etc.). These use a **curated list**, not a live scan of every installed app.
7. Set sleep/wake (stored only; they do **not** auto-block).
8. You land on **Overview**.

Edit setup: Profile tab → **Edit setup** (restarts onboarding). There is no logout.

5.2 Demo A — Timer block (overlay over WhatsApp)

1. Open the **Blocks** tab.
2. Select a duration (e.g. 15 mins).
3. Select **WhatsApp** (or another listed app that is actually installed).
4. Tap **Start Block**. Accept Usage Access + overlay if prompted.
5. Leave Zen Sweep and **open WhatsApp**.
6. You should see a full-screen **cream/forest “WhatsApp is blocked”** overlay with a countdown.
7. **Stop Blocking** or **Return to Zen Sweep** to exit.

If a timer is running, **mood will not show** for that app. Timer always wins.

5.3 Demo B — Mood intervention (no timer)

1. On **Blocks**, turn **Mood Intervention ON** (green switch) for WhatsApp. You do **not** need to start a timer.
2. Confirm a notification like “**Zen Sweep watching**”.
3. Open WhatsApp **without** an active block.
4. You should see “**Take a small pause**” with mood buttons.
5. **Stressed** → breathing counts (inhale / hold / exhale).
Lonely → family-connect prototype.
Other moods → short pause.
6. Tap **Continue** when done.

If mood does not appear: stop any active timer first; wait ~45 seconds after a previous mood Continue (cooldown).

5.4 Other tabs

- **Insights:** local counts of prevented opens and minutes saved; **Share** uses the system share sheet. No server.

- **Overview:** personalized greeting from onboarding.

6. Troubleshooting

Symptom	What to do
Expo Go / "Unable to load" / Dev Client error screen	Metro is not running, or the phone cannot reach it. Start <code>npx expo start</code> in frontend/, then <code>adb reverse tcp:8081 tcp:8081</code> (or 8083). Reopen the app with the <code>exp+frontend://...127.0.0.1...</code> URL.
adb: device offline / install failed	Unplug USB, revoke USB debugging, replug, adb devices. Unlock the phone.
ForegroundServiceDidNotStartInTimeException crash	Fixed in current source. Rebuild with <code>npx expo run:android</code> (JS reload is not enough).
Mood overlay flashes then vanishes	Rebuild with current source (overlay was incorrectly dismissed when UsageStats reported Zen Sweep).
Overlay never appears	Grant Usage access and Display over other apps . Use a physical device . Confirm the app you open is on the list (<code>com.whatsapp</code> , not WhatsApp Business).
Timer overlay instead of mood	Stop the block first. Mood only runs when that package is not timer-blocked.
Native UI (overlay colors/layout) unchanged after edit	Run <code>npx expo run:android</code> again.
JAVA_HOME / Gradle JDK errors	Point JAVA_HOME at Android Studio jbr , not an old Java 8.
Samsung "put this app in deep sleep"	Settings → Battery → Zen Sweep → Unrestricted .
Blocking does nothing on iOS or web	Expected. Core features are Android-only.
Empty app list / missing icons	The picker is a mock list in <code>src/features/appBlocking/services/installedApps.ts</code> . Apps not on that list cannot be selected.
Need a clean profile	Uninstall Zen Sweep or clear app storage (AsyncStorage). Or Profile → Edit setup.

Useful commands

```
cd frontend
npx expo start          # Metro only (after a Dev Client is already installed)
npm run android         # native rebuild + install
npm run typecheck
adb reverse tcp:8081 tcp:8081
adb shell am force-stop com.zensweep.app
```

Quick judge checklist

1. `cd frontend && npm install && npx expo run:android`
2. Complete onboarding (any name + pick WhatsApp).
3. Grant Usage Access + Display over other apps.
4. **Start Block** → open WhatsApp → timer overlay.
5. Stop block → Mood ON → open WhatsApp → mood pause.